

$$\begin{array}{ccccc}
T^2[X, Y] & \xrightarrow{TU^T\psi_{X,Y}} & T[TX, TY] & \xrightarrow{U^T\psi_{TX, TY}} & [T^2X, T^2Y] \\
\downarrow U^T\epsilon_{FT[X,Y]} & & \downarrow U^T\epsilon_{[FTX, FTY]_T} & & \downarrow [T^2X, U^T\epsilon_{FTY}] \\
& & & (iv) & \\
& & & & \\
& & [TX, TY] & \xrightarrow{U^T[\epsilon_{FTX}, FTY]_T} & [T^2X, TY] \\
& \nearrow U^T\psi_{X,Y} & & & \uparrow [U^T\epsilon_{FTX}, TY] \\
T[X, Y] & \xrightarrow{U^T\psi_{X,Y}} & & & [TX, TY]
\end{array}$$

$\epsilon \text{ nat}$ (near the arrow from $T[X, Y]$ to $[TX, TY]$)
 $U^T \text{ closed}$ (near the arrow from $[TX, TY]$ to $[TX, TY]$)